Week 3 Notes & Quiz

Notes for all six Week 3 lectures, made only from the 180 slides in *WEEK 3.pdf*. Each yellow tag such as S29 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L12	**Reverse Logistics**	S1–S16
L13	**Reverse Logistics (contd.)**	S17–S57
L14	**Green Logistics**	S58–S96
L15	**Lean Logistics**	S97–S127
L16	**Lean Logistics (contd.): methodologies and VSM**	S128–S144
L17	**Lean Logistics (contd.): DMAIC, JIT, 5S, Kaizen, SMED, Toyota**	S145–S180
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 12 S1–S16

Reverse Logistics

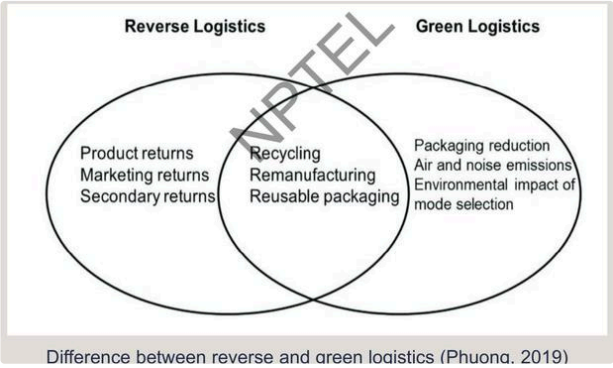
- Reverse logistics is traditional logistics with the flow in the **opposite direction**, moving **waste and returns**. S3
- Other names: retro logistics, reverse supply chain, reverse flow, backward supply chain, returns management, aftermarket logistics, aftermarket supply chain. S3
- Defined as taking waste and returns along the supply chain and reworking them to create a new product that can be sold (Stanisławski & Szymonik, 2023). S3

AUTHOR	DEFINITION S5–S6
Lambert and Stock	"Going the **wrong way on a one way street**" because most shipments flow one way.
Carter and Ellram	The return, **upstream movement** of a good or material from reuse, recycling or disposal, often done by new, auxiliary channel members.
Thierry et al.	Reverse logistics under the umbrella of **product recovery management**.
Rogers and Tibben-Lembke	Adapted from the Council of Logistics Management (below).

> ROGERS AND TIBBEN-LEMBKE · EXACT WORDS
>
> *"The process of planning, implementing, and controlling the efficient, cost-effective flow of raw materials, in-process inventory, finished goods and related information from the point of consumption to the point of origin for the purpose of recapturing value or proper disposal."* S6

Reverse vs forward, reverse vs green

- Forward logistics moves raw materials and components for production; reverse logistics handles everything coming **from the consumer** as waste or return (via recovery facilities, consolidation and collection centres). S7
- Equating reverse logistics with green logistics is **a mistake**: green logistics is about environmentally friendly resources. S8



Difference between reverse and green logistics (Phuong, 2019)

Original slide S8 (Phuong, 2019). Reverse only: product, marketing and secondary returns. Both: recycling, remanufacturing, reusable packaging. Green only: packaging reduction, air and noise emissions, environmental impact of mode selection.

Types of product returns 5 items

TYPE	EXAMPLES (AÏT-KADI ET AL., 2013) S10
Return of unused products	Recovered electronics, recycled carpets, shoes; white and brown goods (Netherlands); ink cartridge collection
Commercial returns	Retailer surplus, clothing, cosmetics, mail-order returns
Product warranty returns	Defective appliances, spare parts

TYPE	EXAMPLES (AÏT-KADI ET AL., 2013) S10
Wastes and derivative products	Pharmaceutical industry, steel components
Packaging	Pallets, crates, bottles; "Green dot system" (Germany)

Why reverse logistics is important

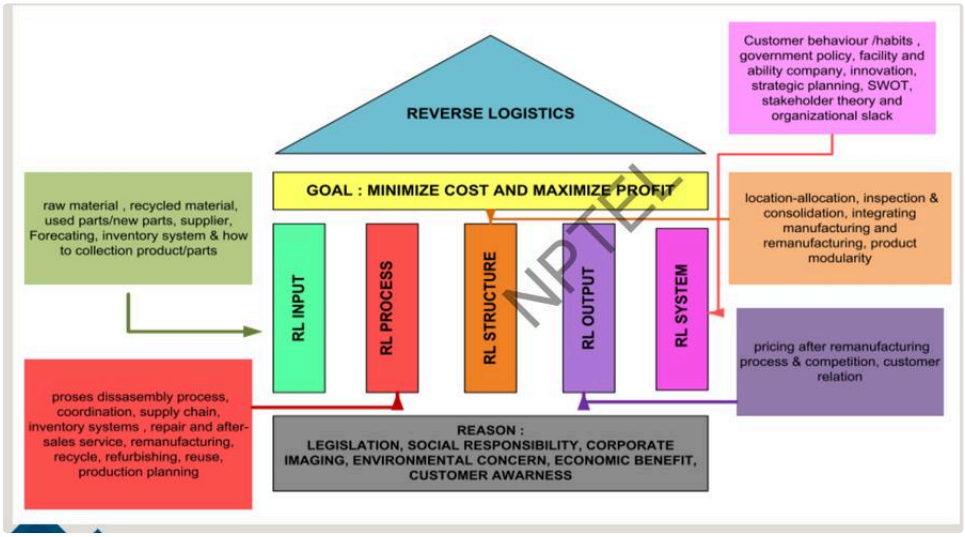
Government policy and legislation	Netherlands (Jan 1999): makers and importers of **white and brown goods** must take back products; similar rule for cars. Germany: **Packaging Ordinance**, firms collect all sales packaging. S11
Economic considerations	US landfill tipping fee rose from **$8 to about $49.78** (1985–2010), "about 600%". Recovery is profitable for car parts, copiers, computers, tyres, aviation equipment. S12
Shift to buying services	Maintenance contracts covering repairs and parts make it easier to take back end-of-life products. S13
Environmental considerations	Cuts cost, ensures sustainability, helps meet legislation, lets firms present themselves as "green". S13
Effective waste management	E-waste: **80% of mobile phones are upgraded every two years**. Sources: households, industries, institutions. Informal recycling common in India and South Africa. S14

LECTURE 13 S17–S57

Reverse Logistics (contd.)

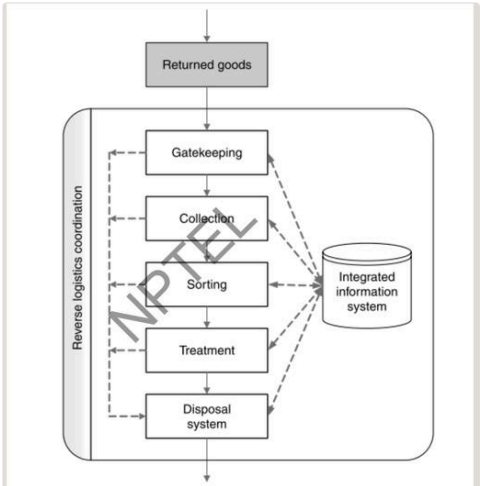
Key aspects 6 items

- **Product returns**: receive, inspect, refurbish, repackage (defects, damage, overstock, dissatisfaction). S19
- **Remanufacturing and refurbishment**: disassemble, repair, clean, reassemble. S19
- **Recycling and disposal**: recycle packaging, extract materials, dispose of hazardous waste responsibly. S20
- **Warranty and service management**. S20
- **Asset recovery and remarketing**: resell, liquidate excess, repurpose. S21
- **Supply chain optimization**: RL software, better routes, integrate with forward logistics. S21



Original slide S23, House of Reverse Logistics (Wang & Sun, 2005). Goal: **minimize cost and maximize profit**. Pillars: RL input, process, structure, output, system. Foundation (reasons): legislation, social responsibility, corporate imaging, environmental concern, economic benefit, customer awareness.

The reverse logistics process



Generic process of reverse logistics (Aït-Kadi et al., 2013)

Original slide S24 (Aït-Kadi et al., 2013): returned goods → gatekeeping → collection → sorting → treatment → disposal system, linked by an integrated information system and reverse logistics coordination.

Gatekeeping	Controls entry into the RL system; rejects returns that don't belong to the business or that it has no contractual or legal obligation for. Leads to **acceptance or rejection**. S25
Collection	Recovery and transport of returned products, by customers, third parties or businesses. S26
Collection case: Canon	Ink cartridge collection program since **1990**; pre-paid shipping coupon in new cartridge packs; now in **24 countries**. S27
Sorting	Validates gatekeeping information; sorts by **condition and reason for return**. S28
Processing / treatment	Options: repackage, repair, disassemble, reconfigure, remanufacture, upgrade, recycle, donate, sell on secondary markets, dispose. **Remanufacturing** restores products to a like-new state and replaces more modules than repair. S29
Disposal	Landfill or incinerator, the **last resort**; remove hazardous materials; incinerators should recover heat for electricity. S30

Factors for success `10 items`

Visibility and transparency · efficient returns management · product recovery and disposition · collaboration with partners · technology and data analytics · reverse logistics network design · compliance and regulation · customer experience · continuous improvement and innovation · sustainability and corporate responsibility. S31–S36

Performance measures `19 items`

Return rate	% of units returned vs sold; high suggests quality or accuracy issues. S38
Return processing time	Time to receive, inspect, process or disposition. S38
Recovery rate	% of returns refurbished, repaired, resold or recycled vs total returns. S39
Disposition cost	Total cost of handling, refurbishing, recycling or disposing. S40
Others	Regulatory compliance, transportation and inventory holding costs, customer education, customer satisfaction, environmental impact, continuous improvement, supplier collaboration, training, technology adoption, risk management, reverse supply chain integration, circular economy initiatives, metrics alignment, stakeholder engagement. S37–S46

Case studies

- **Amazon**: no-fuss returns, RL training programs, full control of a product's life. NRF: a record **$761 billion** of merchandise returned in 2021; **16.6%** of holiday sales returned; online return rate nearly **21%** (18% in 2020); Amazon net sales $469 billion. S47–S48
- Amazon donates eligible returns to charities in North America, Europe, Australia, Japan. S49 SIPP packaging for a Samsung 49" TV removes 9 components, cuts package volume **69%** and air shipped **87%**. S50
- **IKEA** initiatives: generous returns policy, refurbishment centres, material recovery facilities (wood, metal, plastics), circular design (durability, repairability, recyclability). Goal: **carbon neutral by 2030**. S51–S54

LECTURE 14 S58–S96

Green Logistics

- **GSCM** reflects the relationship between the natural environment and supply chain operations, and measures supply chain performance on environmental aspects. S60
- GSCM practices (cycle): green purchasing, production, warehousing, distribution, transportation, raw material, design. S61

Green transportation	Driven by freight CO_2 since the 1990s; up to **71% in 2016**, expected to rise **50% more by 2050**. S62
Green procurement	Buying products and services less harmful to the environment at competitive prices. S62
Green warehouse	Adequate stock levels to avoid material waste and transport cost. S62
Green design	Ecological design: clean, safe factory, lower disposal cost. S63
Green manufacturing	Also **ecological production**: little or no pollution or waste. S63
Green marketing	Also ecological, social, environmental or sustainable marketing. S63

> GREEN LOGISTICS · EXACT WORDS
>
> *"Green logistics, also known as eco-friendly logistics, refers to the practice of managing the flow of goods and resources in a manner that minimizes environmental impact."* S64
>
> Initiatives: alternative fuels, route optimization, energy-efficient vehicles, less packaging waste, reverse logistics for recycling, eco-friendly packaging. S64

- Before the **1960s** there was little concern for environmental degradation; the environment's capacity was seen as infinite (Murphy & Poist, 1995). Green logistics merges research strands from the past 40 years. S65

	GREEN SUPPLY CHAIN	GREEN LOGISTICS
Focus	Sustainability across the **entire** supply chain: design, manufacturing, transport, end-of-life	Also sustainable logistics or eco-logistics; **transportation and distribution** only
Scope	Raw material extraction to final delivery	Emissions, energy use from transport and logistics
Activities	Sustainable sourcing, efficient manufacturing, recycling, green packaging	Route optimization, low-emission vehicles, efficient WMS, sustainable packaging
Objective	Minimize impact of the whole chain while staying profitable	Make logistics operations greener and efficient

Comparison from S67–S69.

Challenges and technologies

- Challenges: rising **raw material prices**; COVID-19 sourcing disruptions; changing management models toward "energy conservation, environmental protection, and low consumption". S70
- Technologies: **green transport** (green vehicle routing counting CO_2-equivalent); logistics solutions for **carbon neutrality** (green procurement, packaging, warehousing; the "last-mile challenge"); **horizontal collaboration** (pooling logistics between two or more manufacturers). S72
- Steps: green solutions; **logistics agglomeration** (clusters) helps cut emissions; carbon management; **decarbonization** = measure emissions, set targets, assess options; alternative fuels for road freight. S73

Green KPIs 9 items

Reverse supply chain assessment	% refurbished or recycled, processing time, cost per unit. S75
Circular economy transition	% recycled or renewable inputs, waste reduction, ROI. S75
Environmental impact & financial performance	Correlation of environmental and financial indicators; fuel efficiency; route deviations. S76
Sustainability & waste management	Waste per unit output, recycling rate, energy efficiency ratio. S77
TOVE	**Transportation Overall Vehicle Effectiveness**: load, fuel, routes; utilization = actual payload ÷ maximum payload. S77
EN 16258 and ISO 14083	International standards for measuring and reporting GHG emissions and energy use in transport. S78
GLEC Framework	**Global Logistics Emissions Council**: calculate and report logistics GHG emissions. S78
Carbon footprint & emission factors	Emissions = activity level × emission factor. S79
Harmonization of emission calculations	Common methods (EN 16258, ISO 14083) and data sharing. S80

SOURCE	EMISSION FACTOR (KG CO_2) S79
Electricity	`kWh/yr × 0.85`
Petrol	`litres/yr × 2.296`
Diesel	`litres/yr × 2.653`

Case studies

- **DHL**: operates in **220 countries**; founded **1969** by Adrian Dalsey, Larry Hillblom and Robert Lynn; sustainability and CSR focus since 2009; **GoGreen** program; aim to cut carbon emissions 50% and pollution 70%; bicycles and electric vehicles for pickup and delivery. S81–S82
- **Myntra**: plastic-free packaging (recycled paper bags, shredded carton instead of bubble wrap, paper tape, Kraft paper invoice pouches, RFID-tagged multi-use polyester bags); **670 tonnes** of plastic diverted by Sept 2021; "e-commerce-ready packaging" (no secondary layer); partnered with **Canopy** and the **Better Cotton Initiative**; two solar-powered fulfilment centres. S83–S85
- **IKEA green transport**: founder's line: any designer can make a $1,000 desk, but a sustainable, beautiful one for **$50** is marvelling. Staff travel plans (bicycles, walking, buses, car sharing); loading shelves replace wooden pallets; transport emissions **19.4%** (2019 report); target **30%** GHG cut from transport in 2020 and 70% by 2030. S86–S87

- Future: **BEVs** suit short city routes; **fuel cell vehicles (FCV)** suit longer routes of large trucks and buses; **DEA** (data envelopment analysis) evaluates green practices; city infrastructure and supplier selection matter. S88–S89

LECTURE 15 S97–S127

Lean Logistics

> LEAN LOGISTICS · PAUL MYERSON (2012)
>
> *"Lean Logistics is an approach to supply chain management that focuses on eliminating waste, improving efficiency, streamlining processes, creating constant flow, enhancing collaboration, and achieving quality."* S100
>
> Alan Harrison (2011): a way to recognise and eliminate wasteful activities to increase product flow and speed. S100

- Benefits: improved efficiency, increased profitability, better customer service, reduced costs. S102
- History: from lean manufacturing in Japan; the term was coined by **John Krafcik** in his **1988** MIT master's thesis; lean manufacturing was pioneered by Toyota engineer **Taiichi Ohno** in the 1970s. S103
- Logistics costs are **5–15% of turnover in industry** and **15–25% in retail**. S104

The 3Ms

Muda · waste	Any activity that adds no value. S106
Muri · overburden	Unnecessary stress on people and processes, caused by Mura and poor training, unclear methods, wrong tools. Leads to breakdowns and absenteeism. S113–S114
Mura · unevenness	Inconsistency. "Mura drives Muda". Example: the month-end rush that produces a hockey-stick production graph. S116–S117

7 types of muda (categorized by Taiichi Ohno in the Toyota Production System): **transport, inventory, motion, waiting, overproduction, over-processing, defects**. S108–S110

- Other wastes: **talent** (unused skills), **resources** (lights and machines left on), **by-products**. Tools to remove waste: Poka-Yoke, Kanban, Takt time, SMED, one-piece flow. S111
- Common waste areas: warehousing ("always wasteful, even if sometimes unavoidable"), trucks (motion), travel time, people, e-vehicles (part of the solution), inventory. S112

Principles of lean logistics 6 items

Eliminating waste · improving efficiency · streamlining processes · creating constant flow · enhancing collaboration · achieving quality. S118–S120

Lean logistics KPIs

- On-time delivery, lead time, inventory turnover, order accuracy, transportation cost per unit, warehouse utilization, cycle time, dock-to-stock time, return rate, order fulfilment cycle time. S122–S123
- **Perfect order rate**: orders delivered with no errors (accurate, complete, intact). **Backorder rate**: orders not fillable immediately. Supplier on-time delivery, order lead time variability, shipment damage rate, sustainability metrics, employee engagement. S124–S125

LECTURE 16 S128–S144

Lean Logistics: Methodologies and Value Stream Mapping

Lean logistics methodologies 13 items

Value stream mapping · JIT inventory · standardized work · continuous improvement culture (Kaizen events) · supplier collaboration · **Lean Six Sigma** (DMAIC, root cause analysis) · technology integration (WMS, TMS) · sustainable packaging · route optimization · demand forecasting · performance-based contracts with LSPs · modular transport (intermodal, containerization) · reverse logistics optimization and performance monitoring. S130–S134

Value stream mapping (VSM)

- Maps the whole value stream from customer request to delivery to find waste and bottlenecks; **defines value from the customer's perspective**. S135
- Deming: "If you can't describe what you are doing as a process, you don't know what you're doing." S135
- Current-state VSM + lean basics = future-state VSM. S136

1. Prepare	Case for change, scope (start and end), requirements, measurements, event logistics. S137
2. Gather data	Develop the current state: walk the value stream, record lead time, cycle time, defects, batch sizes, WIP; map item and information flow. S138
3. Future state & action plans	Future-state map, action plans, timeline. Action plan template: step, problem, person responsible, actions, due date, metrics. S139–S140
4. Execute to plan	Workshops and projects; continuous improvement environment. S141
5. Align	Periodic value stream alignment meetings to review results and update the plan. S142

THE STAPLER EXAMPLE S143–S144

Pick up paper 1 s · walk to stapler 5 s, 20 ft · staple 2 s · walk back 5 s, 20 ft · put down 1 s. **Total 14 s, 40 ft; value-added time 4 s (28%)**, value-added distance 0. The walking is "excess travel".

LECTURE 17 S145–S180

Lean Logistics: DMAIC, JIT, 5S, Kaizen, SMED and Toyota

DMAIC

Define	Team and sponsor agree on scope, goals and financial targets. S148
Measure	Current-state performance, verified with VSM and capability measures. S148
Analyze	Root causes of deviation from customer requirements. S148
Improve	Design improvements, pilot new processes; new KPIs, future-state maps, control plans. S149
Control	Monitoring systems and control plans to sustain results. S149

JIT

- Produce goods **only as needed**: less storage cost, excess inventory and obsolescence risk; needs close supplier coordination. S150
- JIT continuous improvement cycle (8 steps): **Design** (process) → **Manage** (TQM) → **Pull** (Kanban, smaller lots) → **Establish** (vendor relationships) → **Fine-tune** (inventory) → **Build** (skills) → **Refine** (product and process) → **Review** (KPIs, root cause). S151–S153

5S, Kaizen, SMED

- **5S**: Sort (Seiri), Set in order (Seiton), Shine (Seiso), Standardize (Seiketsu), Sustain (Shitsuke). S154–S155
- **Kaizen** = **Kai** (change) + **Zen** (better) = continuous improvement: "Everybody! Everyday! Everywhere!" Small improvements are faster, lower risk, and add up to more than one big change. S156 S173
- Kaizen examples: easy to lubricate, easy to inspect, tyre changeover cut from **5–10 minutes to 5–10 seconds**. S157
- **SMED** (single-minute exchange of die): changeovers in **under 10 minutes** ("single-digit minute"). **OTED** (one-touch exchange of die): under **100 seconds**. S158

Challenges in implementing lean 8 items

Cultural resistance (the biggest) · lack of leadership commitment · silo mentality · limited employee engagement · inadequate training · measurement and accountability · sustainability of gains · external factors. S159–S161

Factor	Lean logistics	Agile logistics
Objective	Efficient operations	Flexibility to meet demands
Method	Remove all waste	Customer satisfaction
Constraint	Customer service	Cost
Rate of change	Long-term stability	Fast reaction to changing circumstances
Measures of performance	Productivity, utilisation	Lead time, service level
Work	Uniform, standardised	Variable, more local control
Control	Formal planning cycles	Less structured by empowered staff

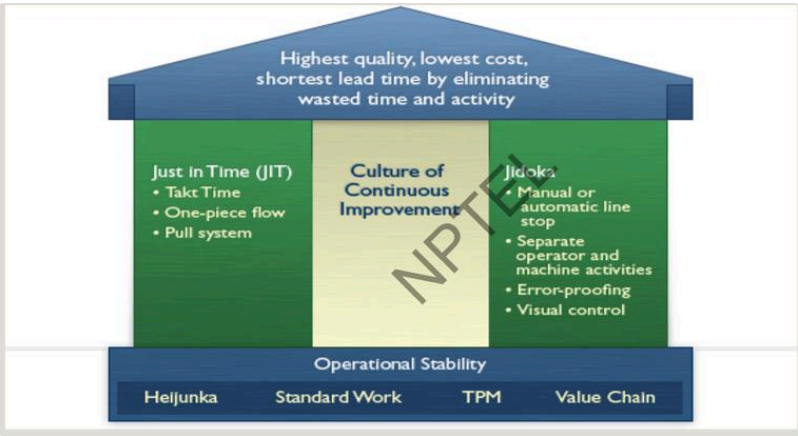
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Source: https://www.creativesafetysupply.com/articles/lean-logistics/

Original slide S162. Lean: efficient operations, remove all waste, constrained by customer service, long-term stability, measured by productivity and utilisation. Agile: flexibility to meet demand, customer satisfaction, constrained by cost, fast reaction, measured by lead time and service level.

Case: Toyota Production System (TPS)

- Toyota: HQ Toyota, Aichi; founder **Kiichiro Toyoda**; TPS founders **Taiichi Ohno** and **Shigeo Shingo**. S163
- 7 TPS principles: reduced setup time, small-lot production, employee involvement, quality at the source, equipment maintenance, pull production, supplier involvement. S164



Original slide S165, the TPS house. Roof: highest quality, lowest cost, shortest lead time. Pillars: **JIT** (takt time, one-piece flow, pull) and **Jidoka** (line stop, separate operator and machine work, error-proofing, visual control), with continuous improvement in the middle. Base: operational stability: heijunka, standard work, TPM, value chain.

- Batch & push: 21 min for the first piece, 30+ min for the order. Continuous flow ("make one, move one"): **3 min** first piece, **12 min** total. S167
- "Just-in-Time" was coined by Kiichiro Toyoda in **1937**; after WWII Taiichi Ohno moved from Toyoda Spinning and Weaving to drive JIT and Jidoka. S168
- **Jidoka** principles: discover an abnormality, stop the process, fix the immediate problem, investigate and solve the root cause. **Poka-yoke**: mistake-proofing; low-cost devices that detect abnormalities or stop the machine before defects are made. S169–S170
- **GEMBA**: Genchi Genbutsu (go and see), Engage (5 whys), Muda/Mura/Muri, Be respectful, Analyze. S171
- **Heijunka**: production smoothing to level fluctuating demand. S172
- Standardized work has three elements: **takt time, working sequence, standard in-process stock**. S174
- Per-head production rose from **39.7 to 70.8** vehicles (about 80%) after TPS. S176
- US market share: GM 40% vs Toyota 2% (1970); Toyota 3% (1980), 8% (1990), 9% (2000), **13% in 2006** while GM fell to 26%. Japanese word **jojo**: "slowly, gradually, and steadily". S178

BEFORE THE EXAM ALL LECTURES

Cheat sheet

NUMBER	MEANING	SLIDE
1999	Netherlands take-back law for white and brown goods	S11
$8 → $49.78	US landfill tipping fee, 1985–2010	S12
80% / 2 yrs	Mobile phones upgraded every two years	S14
1990 · 24	Canon cartridge program start · countries	S27
$761 bn	US merchandise returned in 2021 (NRF)	S48
0.85 · 2.296 · 2.653	Emission factors: electricity · petrol · diesel	S79
670 t	Plastic diverted by Myntra	S84
1988	Krafcik coins "lean" (MIT thesis)	S103
5–15% · 15–25%	Logistics cost share of turnover: industry · retail	S104
28%	Value-added time in the stapler example	S144
<10 min · <100 s	SMED · OTED changeovers	S158
1937	"Just-in-Time" coined by Kiichiro Toyoda	S168
39.7 → 70.8	Toyota vehicles per head after TPS	S176

RL process	Gatekeeping → collection → sorting → treatment → disposal S24
3Ms	Muda (waste), Muri (overburden), Mura (unevenness) S105
7 wastes	Transport, inventory, motion, waiting, overproduction, over-processing, defects S108
VSM steps	Prepare → gather data → future state → execute → align S136
DMAIC	Define, Measure, Analyze, Improve, Control S148
TPS pillars	JIT and Jidoka on a base of operational stability S165

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

The 7 wastes spell **TIM WOOD**: Transport, Inventory, Motion, Waiting, Over-production, Over-processing, Defects.

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Landfill fee maths:** $8 → $49.78 is about a 522% increase (6.2×), but the slide says "about 600%". Quote 600% if asked from the slide. S12
- **Toyota per-head output:** 39.7 → 70.8 is a 78% rise; the slide rounds to 80%. S176
- **SMED** is written "Single-minute Digit exchange of die"; the standard name is single-minute exchange of die. S158
- "Kiichiro Toyota" on S168 is Kiichiro Toyoda (S163 spells it correctly). S168
- IKEA targets: "carbon neutral by 2030" (S53) vs "eliminate the emission of carbon gases to 70% by 2030" (S87). S53 S87
- "Alan Harrrison" (three r's) in the references is Alan Harrison. S100
- Video or link-only slides: S9, S22, S66, S101. S4 has a title only.

SELF-TEST 50 QUESTIONS

Week 3 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 12

Lambert and Stock describe reverse distribution as:

A going the wrong way on a one way street
B waste to energy conversion
C product recovery management
D the return, upstream movement of a good resulting from reuse, recycling or disposal

Q2 LECTURE 12

Rogers and Tibben-Lembke define reverse logistics as the flow from the point of ____ to the point of ____.

A sale; disposal
B production; sale
C consumption; origin
D origin; consumption

Q3 LECTURE 12

Treating reverse logistics as the same as green logistics is, per the slides:

A a mistake
B correct
C true only for packaging
D true only in Europe

Q4 LECTURE 12

In the reverse vs green logistics Venn diagram, which item lies in the overlap of both?

A Product returns
B Packaging reduction
C Remanufacturing
D Air and noise emissions

Q5 LECTURE 12

From January 1999, which country required manufacturers and importers of white and brown goods to take back products after use?

A France
B Netherlands
C Sweden
D Germany

Q6 LECTURE 12

Germany's Packaging Ordinance requires companies to:

A use only paper packaging
B pay a landfill tax
C collect all sales packaging materials
D label recyclable goods

Q7 LECTURE 12

Per the slides, what share of mobile phones are upgraded every two years?

A 80%
B 60%
C 40%
D 95%

Q8 LECTURE 12

'Pallets, crates, bottles' and the 'Green dot system' are examples of which return type?

A Return of unused products
B Packaging
C Product warranty returns
D Commercial returns

Q9 LECTURE 13

In the generic reverse logistics process, which stage comes first after goods are returned?

A Sorting
B Collection
C Treatment
D Gatekeeping

Q10 LECTURE 13

Which stage controls product entry into the reverse logistics system and leads to acceptance or rejection of the return?

A Collection
B Sorting
C Gatekeeping
D Disposal

Q11 LECTURE 13

Canon's ink cartridge collection program began in ____ and now runs in ____ countries.

A 2000; 50
B 1995; 30
C 1990; 24
D 1980; 12

Q12 LECTURE 13

In the reverse logistics process, disposal to landfills or incinerators is treated as:

A the preferred option
B a gatekeeping decision
C the last resort
D the first step

Q13 LECTURE 13

What is the goal at the top of the 'House of Reverse Logistics' (Wang & Sun, 2005)?

A Zero waste
B Minimize cost and maximize profit
C Carbon neutrality
D Customer delight

Q14 LECTURE 13

Remanufacturing differs from repair because it:

A systematically replaces a larger number of used or defective modules to reach a like-new state
B only cleans the product
C sends the product to landfill
D is done only by customers

Q15 LECTURE 13

The 'recovery rate' performance measure is:

A cost of disposing returns
B time to process returns
C % of returned products refurbished, repaired, resold or recycled vs total returns
D % of units returned vs sold

Q16 LECTURE 13

According to the National Retail Federation, how much merchandise was returned to US retailers in 2021?

A $469 billion
B $761 billion
C $561 billion
D $1 trillion

Q17 LECTURE 13

SIPP packaging for a Samsung 49" TV reduced package volume by:

A 9%
B 49%
C 69%
D 87%

Q18 LECTURE 14

Green logistics is also known as:

A lean logistics
B agile logistics
C eco-friendly logistics
D reverse logistics

Q19 LECTURE 14

Green logistics, compared with green supply chain management, focuses mainly on:

A marketing
B product design
C transportation and distribution
D raw material extraction

Q20 LECTURE 14

Green manufacturing is also known as:

A ecological production
B social marketing
C green procurement
D horizontal collaboration

Q21 LECTURE 14

Pooling logistics activities and consolidating supply chains between two or more manufacturers is called:

A horizontal collaboration
B decarbonization
C logistics agglomeration
D vertical integration

Q22 LECTURE 14

What does TOVE stand for?

A Total Output per Vehicle Engine
B Transport Optimization and Vehicle Efficiency
C Total Operating Vehicle Emissions
D Transportation Overall Vehicle Effectiveness

Q23 LECTURE 14

Which pair are international standards for measuring and reporting GHG emissions in transport operations?

A EN 16258 and ISO 14083
B GLEC and GRI
C ISO 26000 and EN 1090
D ISO 9001 and ISO 14001

Q24 LECTURE 14

A vehicle uses 1,000 litres of diesel a year. Using the slide's emission factor, its emissions are about:

A 2,653 kg CO_2
B 3,000 kg CO_2
C 850 kg CO_2
D 2,296 kg CO_2

Q25 LECTURE 14

DHL's environmental program is called:

A Better Cotton
B Canopy
C SIPP
D GoGreen

Q26 LECTURE 14

For short city routes the slides say the optimal green vehicle is the:

A diesel hybrid
B battery electric vehicle (BEV)
C CNG truck
D fuel cell vehicle (FCV)

Q27 LECTURE 15

The term 'lean' was coined by ____ in his 1988 master's thesis at MIT.

A Taiichi Ohno
B Paul Myerson
C Shigeo Shingo
D John Krafcik

Q28 LECTURE 15

Logistics expenses typically account for what share of turnover in retail?

A 5–15%
B 1–5%
C 15–25%
D 25–40%

Q29 LECTURE 15

Muri means:

A continuous improvement
B unevenness
C waste
D overburden

Q30 LECTURE 15

Which statement do the slides make about Mura?

A Muda drives Muri
B Muri drives Mura
C Mura is unrelated to waste
D Mura drives Muda

Q31 LECTURE 15

Which is NOT one of Taiichi Ohno's seven types of muda?

A Talent
B Waiting
C Over-processing
D Motion

Q32 LECTURE 15

The month-end rush that leaves the first week of the next month slow (a 'hockey stick' graph) illustrates:

A Muri
B Muda
C Kaizen
D Mura

Q33 LECTURE 15

Which is NOT listed as a principle of lean logistics?

A Enhancing collaboration
B Maximizing safety stock
C Creating constant flow
D Achieving quality

Q34 LECTURE 15

'Percentage of orders delivered with no errors, complete and intact' is the:

A shipment damage rate
B perfect order rate
C order lead time variability
D backorder rate

Q35 LECTURE 16

Value stream mapping defines value from whose perspective?

A The regulator's
B The supplier's
C The shareholder's
D The customer's

Q36 LECTURE 16

What are the five steps of the value stream process in order?

A Prepare, Gather data, Future state & action plans, Execute to plan, Align
B Define, Measure, Analyze, Improve, Control
C Plan, Do, Check, Act, Review
D Sort, Set, Shine, Standardize, Sustain

Q37 LECTURE 16

In the stapler example, total time is 14 seconds and value-added time is 4 seconds. Value-added time is about:

A 14%
B 28%
C 72%
D 40%

Q38 LECTURE 16

Lean Six Sigma uses which structured technique?

A SMED
B FAST
C GEMBA
D DMAIC

Q39 LECTURE 16

Tying LSP payment terms to on-time delivery, order accuracy and inventory accuracy is the methodology called:

A Modular transportation
B Supplier collaboration
C Route optimization
D Performance-based contracts

Q40 LECTURE 16

Which quote appears on the VSM slide?

A 'Fit for use' (Juran)
B 'What gets measured gets managed' (Drucker)
C 'Quality is free' (Crosby)
D 'If you can't describe what you are doing as a process, you don't know what you're doing' (Deming)

Q41 LECTURE 17

In DMAIC, which phase establishes the current-state performance of a process?

A Define
B Analyze
C Measure
D Control

Q42 LECTURE 17

In the JIT continuous improvement cycle, which step uses signalling methods like Kanban?

A Manage
B Design
C Refine
D Pull

Q43 LECTURE 17

'Kaizen' combines Kai and Zen, meaning:

A go + see
B waste + removal
C change + better
D stop + fix

Q44 LECTURE 17

SMED aims for changeovers of under ____, while OTED aims for under ____.

- **A** 1 minute; 10 seconds
- **B** 100 seconds; 10 seconds
- **C** 60 minutes; 10 minutes
- **D** 10 minutes; 100 seconds

Q45 LECTURE 17

According to the slides, the biggest challenge in implementing lean is often:

- **A** supplier capacity
- **B** cultural resistance
- **C** external factors
- **D** technology cost

Q46 LECTURE 17

In the lean vs agile table, agile logistics is measured mainly by:

- **A** productivity and utilisation
- **B** lead time and service level
- **C** uniform standardised work
- **D** formal planning cycles

Q47 LECTURE 17

The two pillars of the Toyota Production System house are:

- **A** Kanban and Andon
- **B** Heijunka and TPM
- **C** Kaizen and 5S
- **D** Just-in-Time and Jidoka

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	Lambert and Stock: 'going the wrong way on a one way street'. The upstream-movement definition is Carter and Ellram's. S5
Q2	C	From the point of consumption to the point of origin, to recapture value or dispose properly. S6
Q3	A	S8: identifying reverse logistics with green logistics is a mistake. S8
Q4	C	Overlap: recycling, remanufacturing, reusable packaging. S8
Q5	B	S11: the Netherlands. Germany's example is the Packaging Ordinance. S11
Q6	C	S11. S11
Q7	A	S14: 80%, a major source of e-waste. S14
Q8	B	S10 table, Aït-Kadi et al. (2013). S10
Q9	D	Returned goods → gatekeeping → collection → sorting → treatment → disposal. S24
Q10	C	S25: gatekeeping. S25
Q11	C	S27: started 1990 with a pre-paid shipping coupon; now 24 countries. S27
Q12	C	S30: disposal is used as a last resort. S30
Q13	B	Goal: minimize cost and maximize profit. S23
Q14	A	S29. S29
Q15	C	S39. Return rate (S38) is returns vs units sold. S39
Q16	B	S48: a record $761 billion. $469 billion is Amazon's net sales. S48
Q17	C	69% less volume, 87% less air shipped, 9 fewer components. S50
Q18	C	S64. (S67 adds 'sustainable logistics or eco-logistics'.) S64
Q19	C	GSCM covers the entire chain; green logistics focuses on transport and distribution. S67
Q20	A	S63. S63
Q21	A	S72: horizontal collaboration in green logistics. S72
Q22	D	S77: Transportation Overall Vehicle Effectiveness. S77
Q23	A	S78: EN 16258 and ISO 14083. S78
Q24	A	Diesel factor 2.653 → 1,000 × 2.653 = 2,653 kg CO_2. Petrol is 2.296; electricity 0.85 per kWh. S79
Q25	D	S81: GoGreen; DHL operates in 220 countries and was founded in 1969. S81
Q26	B	BEV suits short city routes; FCV suits longer routes of large trucks and buses. S88
Q27	D	S103: John Krafcik, 1988. S103
Q28	C	5–15% in industry, 15–25% in retail. S104
Q29	D	Muda = waste, Muri = overburden, Mura = unevenness. S113
Q30	D	S116: 'Mura drives Muda!' S116
Q31	A	Talent is listed under 'other wastes' (S111), not the original seven. S108–S111
Q32	D	S117: an example of Mura, unevenness. S117
Q33	B	Principles: eliminate waste, improve efficiency, streamline, constant flow, collaboration, quality. S118–S120
Q34	B	S124: perfect order rate. S124
Q35	D	S135. S135
Q36	A	Prepare → gather data (current state) → future state & action plans → execute → align. S136–S142
Q37	B	4/14 ≈ 28%. Total distance is 40 ft; value-added distance is 0. S144
Q38	D	S131: DMAIC (Define, Measure, Analyze, Improve, Control) and root cause analysis. S131
Q39	D	S133. S133
Q40	D	S135: W. Edwards Deming. S135
Q41	C	S148: Measure, verified through VSM and capability measures. S148
Q42	D	S152: Pull: educate on production and withdrawal methods using Kanban; reduce lot sizes. S152
Q43	C	Kai = change, Zen = better → continuous improvement. S173
Q44	D	S158: single-digit minutes (<10 min); one-touch exchange of die <100 s. S158
Q45	B	S159: one of the biggest challenges is resistance to change. S159
Q46	B	Lean: productivity, utilisation. Agile: lead time, service level. S162
Q47	D	JIT and Jidoka, with continuous improvement between them and operational stability below. S165
Q48	B	S170: mistake proofing or error proofing. Heijunka is production smoothing. S170
Q49	A	S174. S174

Q	ANS	WHY · SLIDE
Q50	C	S168 (spelled 'Kiichiro Toyota' there); the founder of Toyota Motor Corporation. S168

Made from *WEEK 3.pdf* (180 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".